String Instructions

* أي (Data) تخزن في الـ (memory) هي عبارة عن (String)

MOVSB

→ Copy byte at (DS:[SI]) to (ES:[DI])
→ update (SI) and (DI)
→ flag unchanged

→ Algorithm :-

ES:[DI] = DS:[SI]
IF DF = 0 then
SI = SI + 1
DI = DI + 1
else (DF=1)
SI = SI - 1
DI = DI - 1

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REP

* تقوم بإعادة تنفيذ التعليمات التي تليها

→ Repeat following

(MOVSB , MOVSW , LODSB , LODSW , STOSB , STOSW) instructions

→ effect flag

→ Algorithm :-

check cx :
if cx <>0 then
do following chain instruction
cx = cx-1
go back to check cx
else
exit from REP cycle

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MOVSW

→ copy word at (DS : [SI]) to (ES : [DI])

→ update (SI) and (DI)

→ flags unchanged

→ Algorithm :-

```
ES : [DI] = DS : [SI]
IF DF = 0 then
SI = SI + 2
DI = DI + 2
else
SI = SI - 2
DI = DI - 2
```

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STOSB

→ Store byte in (AL) into (ES : [DI])

→ Update (DI)

→ flags unchanged

→ Algorithm :

ES : [DI] = AL
if DF = 0 then
DI = DI + 1
else
DI = DI - 1

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STOSW

→ Store word in (AX) in to (Es : [DI])

→ update (DI)

→ Flags unchanged

→ Algorithm :-

Es : [DI] = AX
if DF = 0 then
DI = DI + 2
else
DI = DI - 2

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LODSB

→ Load byte at (DS:[SI]) in to (AL)

→ update (SI)

→ flags unchanged

→ Algorithm :-

AL = DS:[SI]
If DF=0 then
SI = SI+1
else
SI = SI-1

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LODSW

→ Load word at (DS:[SI]) in to (AX)

→ update (SI)

→ flags unchanged

→ Algorithm :-

```
AX = DS:[SI]
if DF = 0 then
    SI = SI + 2
else
    SI = SI - 2
```

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SCASB

→ compare bytes [AL] from [Es : [DI]]

→ effects on flags

→ Algorithm :-

AL - Es : [DI]
set flags according to result : (OF, SF, AF, PF, CF)
if DF = 0 then
DI = DI + 1
else
DI = DI - 1

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SCASW

→ compare word (AX) from (ES:[DI])

→ effects on flags

→ Algorithm :-

AX - ES[DI]
set flags according to result OF, SF, ZF, AF, PF, CF
if DF = 0 then
DI = DI + 2
else
DI = DI - 2

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CMPSB

→ Compare bytes (Es:[DI]) from (DS:[SI])

→ effects on flags

→ Algorithm :-

DS:[DI] - Es:[DI]
set flags according to result : OF, SF, ZF, AF, PF, CF
if DF=0 then
SI = SI + 1
DI = DI + 1
else
SI = SI - 1
DI = DI - 1

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CMPSW

→ compare words : (ES : [DI]) from (DS : [SI])

→ effect flags

→ Algorithm

DS : [SI] - ES:[DI]
set flags according to result : (OF, SF, ZF, AF, PF, CF)
if DF=0 then
SI = SI+2
DI = DI+2
else
SI = SI-2
DI = DI-2

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REPNE

→ Repeat following (CMPSB / CMPSW / SCASB / SCASW) instructions with (ZF=0) (result is not equal), maximum (CX) times

→ Algorithm :-

→ check CX
→ if CX <>0 then
→ do following chain instruction
→ CX = CX-1
→ if ZF =0 then
→ go back to check CX
→ else
→ exit from REPNE cycle
→ else
→ exit from REPNE cycle

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REPNZ

→ Repeat following (CMPSB / CMPSW / SCASB / SCASW) instruction while (ZF=0) (result not equal), maximum (CX) times

→ Algorithm :-

→ check CX
→ if CX <>0 then
→ do following chain instruction
→ CX = CX-1
→ if ZF =0 then
→ go back to check CX
→ else
→ exit from REPNZ cycle
→ else
→ exit from REPNZ cycle

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REPZ

→ Repeat following (CMPSB, CMPSW, SCASB, SCASW) instruction while (ZF=1) (result is zero), maximum (CX) times

→ Algorithm :-

→ check CX
→ if CX <> 0 then
→ do following chain instruction
→ CX = CX-1
→ if ZF=1 then
→ go back to check CX
→ else
→ exite from REPZ cycle
→ else
→ exite from REPZ cycle

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REPE

→ Repeat following (CMPSB , CMPSW , SCASB , SCASW) instructions while (ZF = 1) (result is equal) , maximum (CX) times

→ Algorithm :-

→ check CX
→ if CX <>0 then
→ do following chain instruction
→ CX = CX - 1
→ if ZF = 1 then
→ go back to check CX
→ else
→ exit from REPE cycle
→ else
→ exit from REPE cycle

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